

Response to Reviewers 2

CLDY-D-25-00309: Lorenz Energy Cycle Climatology for the Southwestern Atlantic Cyclones

Response to Reviewer 1

I thank the authors for their detailed and thorough response to my first review. The revised manuscript is in general much clearer and easier to follow. I have a few very minor comments which would further improve the manuscript which are listed below.

R: We appreciate the reviewer's positive assessment and thank you for the careful, constructive comments. We address each minor point below and have made the corresponding revisions in the manuscript.

1. Line 157, the abbreviations for the regions are not explained. Here Figure 1 should be referred to but it would also help a reader if you explained what these abbreviations / shorten terms refer to.

R: We have defined all regional abbreviations at first mention and added an explicit cross-reference to Fig. 1 (L158-158).

2. Section 2.1. This follows up on my previous comment #6 about the units of the cyclone track density. Please can you add some brief details about how this is computed to this manuscript and highlight that it is not the same as often used by TRACK.

R: We have added a brief description of how the track density is computed in our study and clarified how this differs from the TRACK convention (L162-169).

3. Line 212, after "as residuals" could the symbols be added to make it clear what RKZ, RKE, RGZ and RGE are?

R: We have added the explicit symbols to clarify the notation (L221-224).

4. Section 3.4. The name of this sub section could be more informative.

R: We have updated it to "Characteristics of Cyclones Based on EOF Signal".

5. Line 530-531. I am not sure what Figure this statement is referring to. Please add a figure reference. However, if this is Figure 9, I do not agree with the statement that all cyclone tracks extend southeastward across the Atlantic as Figure 9c shows almost a northeastward Track.

R: We have added references to Figures 9 and 10, and clarified that EOF3(+) (Fig. 9c) is an exception (L540-542). Its northeastward displacement is mentioned on L529-531. Thank you for pointing that out.

6. Paragraph starting at line 586: Please refer to specific panels of Figure 14 more in this Paragraph.

R: We have now referred to specific panels of Figure 14 throughout the paragraph for clarity (L596-609). Thank you for the suggestion.

7. Line 636, “latitudes of larger background APE” – suggest adding “climatological” to this sentence.

R: We have done that (L639). Thanks for the suggestion.

8. Line 774 “..for 7531 cyclonic systems”. Please check if this number is accurate as it differs what is specified in line 165 (6789 cyclones).

R: We have changed it to the correct number, which is 6,789 (L777). Thanks for the careful examination.

9. Appendix, equations 17 and 18. There are unbalanced square parentheses in both of these equations – there is either a missing [or an extra].

R: We have corrected the unbalanced square brackets in equations 17 and 18. Thank you for the careful examination.

10. Appendix, line 843, Could the levels that were used for pb

R: We chose not to specify the exact levels in the appendix as the formula can be adapted for different uses. However, in Section 2.1, we mention that we use levels from 1000 to 1 hPa (L132-133).